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Exercise sheet 03, Solution hints
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Homework:

Exercise 1:

Let $p \geq 0$ and $n = 2^p$. A matrix $M \in \mathbb{R}^{2^p \times 2^p}$ has the $\mathcal{H}_p$ format if

1. $M \in \mathbb{R}^{1\times 1}$ for $p = 0$,
2. $M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}$ with $M_{11}, M_{22} \in \mathcal{H}_{p-1}$ and $M_{12}, M_{21} \in R(k, 2^{p-1}, 2^{p-1})$.

Proof that the number of matrix blocks $N_{\text{block}}(p)$ of an $\mathcal{H}_p$-matrix is given by $3n - 2$.
Solution hints: Proof by induction:

Base case: Let $p = 0$. Then $N_{\text{block}}(p) = 1$.
Induction hypothesis: For $p > 0$, we assume $N_{\text{block}}(p-1) = 3 \cdot 2^{p-1} - 2$. Induction step: With the induction hypthesis, it follows

$$N_{\text{block}}(p) = \underbrace{2}_{\mathcal{R}_{p-1}-\text{blocks}} + \underbrace{2}_{\mathcal{H}_{p-1}-\text{blocks}} \cdot N_{\text{block}}(p-1) = 2 + 2 \cdot \left(3 \cdot 2^{p-1} - 2\right) = 3 \cdot 2^p - 2 = 3n - 2.$$

Exercise 2: Let $M = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}$ with nonsingular block M_{11} and Schur complement $S = M_{22} - M_{21}M_{11}^{-1}M_{12}$. Show that:

a) M is nonsingular if and only if S is nonsingular.

b) If M is symmetric positive definite, then S is symmetric positive definite as well.

Solution hints:

a) With

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} = \begin{bmatrix} I & 0 \\ M_{21}M_{11}^{-1} & I \end{bmatrix} \begin{bmatrix} M_{11} & M_{12} \\ 0 & S \end{bmatrix}$$

and

$$\det M = \det \begin{bmatrix} I & 0 \\ M_{21}M_{11}^{-1} & I \end{bmatrix} \cdot \det \begin{bmatrix} M_{11} & M_{12} \\ 0 & S \end{bmatrix} = \det M_{11} \cdot \det S$$

we find M nonsingular $\Leftrightarrow$ S nonsingular.

b) If M is symmetric, then M_{11}, M_{22} are symmetric and we have $M_{21} = M_{12}^T$. It follows that S is symmetric as well. Let $M \in \mathbb{R}^{n,n}$ be symmetric positive definite (SPD) with $M_{22} \in \mathbb{R}^{m,m}$. With

$$M = \underbrace{\begin{bmatrix} I & 0 \\ M_{21}M_{11}^{-1} & I \end{bmatrix}}_{=:L} \begin{bmatrix} M_{11} & 0 \\ 0 & S \end{bmatrix} \underbrace{\begin{bmatrix} I & M_{11}^{-1}M_{12} \\ 0 & I \end{bmatrix}}_{=L^T,\ \text{wenn } M_{12}=M_{21}^T}$$

follows that

$$L^{-1}M(L^{-1})^T = \begin{bmatrix} M_{11} & 0 \\ 0 & S \end{bmatrix}.$$

For an arbitrary $x \in \mathbb{R}^m \setminus \{0\}$, we define $y = (0^T \;\; x^T)^T \in \mathbb{R}^n$. We obtain

$$x^T S x = y^T \begin{bmatrix} M_{11} & 0 \\ 0 & S \end{bmatrix} y = y^T L^{-1} M (L^{-1})^T y > 0$$

since M is SPD and $(L^{-1})^T y \neq 0$.

Exercises (in class):

Exercise 3: An $\mathcal{H}_p$-matrix is stored in the format `supermatrix` that also allows more general formats but comprises the $\mathcal{H}_p$-format as a special case.

```
typedef struct _supermatrix supermatrix;
typedef supermatrix *psupermatrix;
typedef const supermatrix *pcsupermatrix;

struct _supermatrix {
  int block_rows;
  int block_cols;
  int rows;
  int cols;
  prkmatrix r;
  pfullmatrix f;
  psupermatrix *s;
};
```

Every `rows`×`cols` `supermatrix` has a tensor block structure where the `block_rows`×`block_cols` subblocks are stored in one of the formats `fullmatrix`, `rkmatrix` or `supermatrix`. A `supermatrix`-block is:

a) a `fullmatrix` if `f` $\neq$ NULL and `r` = NULL, `s` = NULL,

b) a `rkmatrix` if `r` $\neq$ NULL and `f` = NULL, `s` = NULL,

c) a `supermatrix` if `s` $\neq$ NULL and `r` = NULL, `f` = NULL.

For the cases a) and b), it holds that `block_rows` = `block_cols` = 1. For the case c), the subblocks are sorted columnwise.
For a given $p > 0$, if holds for a $\mathcal{H}_p$-matrix

$$\texttt{rows} = \texttt{cols} = 2^p,\ \texttt{block_rows} = \texttt{block_cols} = 2,\ \mathrm{r} = \mathrm{f} = \texttt{NULL},\ \mathrm{s} \neq \texttt{NULL}.$$

Furthermore, `s[0]` and `s[3]` are then matrices in the `supermatrix`-format with analogous structure and $p-1$ ($\mathcal{H}_{p-1}$-matrices). `s[1]` and `s[2]` are matrices in the `rkmatrix`-format ($\mathcal{R}_{p-1}$-matrices).
Familiarize yourself with the methods in *supermatrix.h*, *supermatrix.c*. Implement the following methods (in *supermatrix.c*):

- `supermatrix new_hp_matrix(int p);`

  for creating the structure of a $\mathcal{H}_p$-matrix,

- `void addeval_hp_supermatrix(pcsupermatrix s, const double *v, double *w);`

  for the matrix-vector product of the form $\texttt{w} := \texttt{w} + \texttt{S} * \texttt{v}$ with a $\mathcal{H}_p$-matrix `S`.

Test your methods for the matrix-vector products with *example_ 03.c*. Can you confirm the theoretical complexity of $\mathcal{O}(n \log n)$ for the matrix-vector multiplication for $\mathcal{H}_p$-matrices?
Solution hints:

Exercise 4: The method

```
size_t getsize_supermatrix(pcsupermatrix s);
```

calculates the memory complexity in bytes that is needed for a matrix in the format `supermatrix`. Use this method to verify the theoretical memory complexity of $\mathcal{O}(n \log n)$ for a $\mathcal{H}_p$-matrix in the format `supermatrix`. Add your code to the part of *example_ 03.c* marked with `/* EXERCISE */`.
Solution hints: